

Reza Mirbagheri

Mechatronics Engineer

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Education

2023 - Present

Master of Science in Mechatronics engineering
Tor vergata University, Rome, Italy

2013 – 2018:

Bachelor of Science in Mechanical engineering
Azad University, Karaj, Iran

Thesis: Water tank level control using LPV (Linear Parameter Varying)
implementation with MATLAB

Supervisor: Dr. Abolfath Nikranjbar

Complementary Education

2022/06 – 2022/09:

4th National Robotics and Mechatronics Workshop
Iran Association of Robotics and Mechatronics

Courses taken:

(Arduino – Robots design with SOLIDWORKS – Proteus – Python
Programming & Machine Learning – PCB design – Robotics teaching
method)

2022/06 – 2022/09:

Electronics and Control in Robotics Internship
K. N. Toosi University

ARAS (Advanced Robotics and Automated System) Research Institute

Courses taken:

STM32 Microcontroller simulation with Proteus and Implementation
of a PID Controller

Summer 2021:

Machine Learning Summer School (MLSS 2021)

NTU (National Taiwan University)

Courses taken:

Reinforcement learning – Theory of Deep learning

Research Interests

- Optimal and Intelligent Control
- Robotics and Intelligent Systems
- Mechatronics and Automation Systems
- IOT and Cyber-Physical Systems
- AI, Machine/Reinforcement Learning
- Embedded and Embedded Linux

Work Experience

11/2022 – Present

Python developer

HeyvaAI, Tehran, Iran

Implemented **RESTful APIs** using Flask, enabling seamless data exchange between the cryptocurrency platform's front-end and blockchain services.

09/2019 – 06-2020

Teacher, Middle and high school students

Naser Khosro High School, Karaj, Iran

Taught **Vocational Technology** to students in grades **7, 8, and 9**.

06/2017 – 06/2018:

Educational Counselor (sophomore students)

Ghalamchi Cultural Education Center, Karaj, Iran

Served as an **Educational Counselor** for 10th-grade students, providing guidance on **time management** and **effective study techniques**.

Skills

- **Programming Language:** Python, MATLAB, C++
- **Back-end Framework:** FastAPI
- **Software:** Simulink, PVsyst (Solar Panels) – MESHMIXER (3D printing) – Proteus – Altium Designer
- **Computer-aided Design (CAD):** SOLIDWORKS, Autodesk Inventor
- **Machine Learning:** Python (NumPy – Pandas – Scikit-learn)
- **Microcontroller:** STM32 - Arduino - ESP32

Projects

- Control the temperature and light of the environment using LDR, NTC sensors with Arduino and Proteus
- Design Combat robots with SOLIDWORKS
- Setting up electronics circuits with STM32 Microcontroller
- Design, Analyses, and Prototype rehabilitation instruments by CATIA, MESHMIXER software, and 3D printer to help people who can't use their hands to pick up cups, spoon, fork, cellphone, and variety of different products.
- Design an Off-grid and On-grid solar farm using PVsyst software.

Certificates

- **Electronics and Control in Robotics Internship: 2022**
K. N. Toosi University (ARAS)
- **4th National Robotics and Mechatronics Workshop: 2022** in Fourth National Robotics and Mechatronics Workshop
Iran Association of Robotics and Mechatronics
- **Machine Learning Summer School: 2021**
NTU (National Taiwan University)
- **Machine Learning with Python: 2021**
IBM (Coursera)
- **Speak English Professionally: In Person, Online & On the Phone: 2021**
Georgia Institute of Technology (Coursera)
- **Getting Started with Essay Writing: 2021**
UCI (Division of Continuing Education) (Coursera)

- **Photovoltaic Systems Designer (Renewable & Alternative Energies Technology):**
07/2019
Iran Technical and Vocational Training Organization
- **Complex & Freeform surfaces using CATIA Software (Advanced):** 06/2019
Iran Technical and Vocational Training Organization
- **Rapid Prototyping with 3D printing method:** 01/2019
Iran Technical and Vocational Training Organization
- **Hydrodynamic Power Transmission System Repairman:** 10/2018
Iran Technical and Vocational Training Organization
- **CATIA Operator:** 08/2018
Iran Technical and Vocational Training Organization
- **NANO Technology General Training:** 2013
Chamran Nanotechnology Research Center

Language & Test Scores

- **English:** Fluent
- **Korean:** A2-Studied in King Sejong Institute, Tehran
- **Italian:** A1
- **Persian:** Native